Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/04/2005
PI Name:	Faghri, Mohammad Ph.D.		
Project Title:	Phase Change in Low and Jittering Gravity Environment Simulated via Electromagnetic Field		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02881-0805	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	05/30/2000	**End Date:**	11/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Faghri, M.		
COI Name (Institution):	Charmchi, M. (Univ. of Massachusetts-Lowell) Kalluri, D. K. M. (University of Massachusetts-Lowell)		
Grant/Contract No.:	NAG3-2434		
Performance Goal No.:			
Performance Goal Text:			
	Transport processes associated with melting of an electrically conducting Phase Change Material (PCM), placed inside a rectangular enclosure, under low-gravity environment, and in the presence of an electromagnetic field is simulated numerically and experimentally. Electromagnetic forces damp the natural convection as well as the flow induced by sedimentation and/or floatation, and thereby simulating the low gravity environment of outer space. Computational experiments are conducted for both sidewall heating and top-wall heating in both two- and three-dimensional rectangular enclosures. The governing equations are discretized using a control-volume-based finite difference scheme. Numerical solutions are obtained for true low-gravity environment as well as for the simulated-low-gravity conditions. The effects of electromagnetic field on the natural convection, solid phase floatation/sedimentation, liquid-solid interface location, solid melting rate, and flow patterns are investigated. It is found that the melting under low-gravity environment can reasonably be simulated on earth via applying a strong horizontal magnetic field. However, the flow		

Task Description:	patterns obtained for the true low-gravity cases are not similar to the corresponding cases solved for the simulated-low-gravity environment.On the other hand, it is shown that under electormagnetic field simulation, where the Lorentz force is caused by the transverse electric and magnetic fields, it is possible to minimize the flow field distortiion caused by the high magnetic field and therefore achieving a much better simulation of low-gravity. Both solidification and melting experiments were performed. These set of experiments are for sidewall phase change processes and therefore, addresses the fixed solid phase conditions. The effects of magnetic field on phase change rate and on the shape of the solid/melt interface are studied. The solid thickness is measured via ultrasonic techniques and the solid/melt interface is mapped using florescent light shadowgraphy through a transparent window. The collected data consist of temperature history, ultrasonic detection of the interface, florescent light shadowgraphy and solid phase volume fraction. The results showed that the presence of magnetic field had a marked effect on melting and natural convection whereas; phase change convection was noticeable in the solidification cases. The objective of the experimental part of the project is to design the Experimental Apparatus and Measurement System. Experiments will be conducted to observe solidification and melting processes under low gravity condditions induced by electromagnetic fields and to measure the temperature field and the solid thickness as a function of time and wall temperature conditions. The objective of the numerical part of the project is to model and solve the phase change problem under low gravity environment in two- and three-dimensional enclosure. Also, preliminary computations will be conducted to assist with the design of the experiment. Tasks Performed in the Numerical Part A (M. Faghri)Development of the computer codeValidating the code in the limiting casesFormulation of the ProblemSimulation of some cases to assist experiments. Also conduct computational experiments for both fixed(sidewall heating)and unfixed(top-wall heating)PCM in three-dimensional enclsoures. Tasks Performed in the Experiment Part B (M. Charmchi)Review of experiment design requirements: test cell dimensions, materials selections, boundary conditions, fabrication procedures, and assembly considerations.Instrumentation Requirements: data requirements and data acquisitions:Temperature measurements (sensors specifications and positions)Thermal environmental control (power requirements -heating and cooling- and equipment specifications)Data acquisition system (I/O boards, PC interface, and DAQ software). Conduct experiments to observe solidification and melting processes under low gravity conditions simulated by electromagnetic fields and measure temperature field and solid thickness as a funciton of time and the wall temperature conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	3D-Phase Change Simulated Low Gravity via Electromagnetic Fields (Sidewall): o Received paper with reviewer comments for International Journal of Numerical Methods in Heat and Fluid Flow. o Implemented reviewer comments, which improved the paper and resubmitted for publication. o This paper was accepted for publication in the journal above due to be published in 2005. 3D-Phase Change Simulated Low Gravity via Electromagnetic Fields (Topwall): o Completed mathematical formulation in three dimensions o Implemented formulation into the CFD code. o Validate the CFD code for the Topwall problem. o Completed a parametric study. o Completed write up of Topwall paper. o Received internal reviewer comments and implemented suggestions. o Submitted the Topwall Paper for publication in Numerical Heat Transfer. 3D-Phase Change Simulated Low and Jittering Gravity via Electromagnetic Fields (Sidewall): o Completed mathematical formulation in three dimensions to include for the first time the induced magnetic fields and surface tension o Implemented formulation into the CFD code. o Validated CFD code for this paper. o Completed grid study for this problem. o Began parametric study, currently underway. This problem will be completed and submitted for publication by the end of 2005. Numerical CFD code validation vs. UMASS-Lowell Sidewall heated experiment o The experimental results of Charmchi et al. have been used to validate the numerical CFD code. o Good correlation has been achieved between the case with and without the application of magnetic fields.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Articles in Peer-reviewed Journals	Goncalves, E., Faghri, M., Asako, Y. and Charmchi, M. "Numerical Solution of Melting Processes For Unfixed Phase-Change Material in the Presence of Electromagneticallly Simulated Low Gravity" Numerical Heat Transfer, Part A, Vol. 46, pp. 343-365 , Apr-2004
Awards	Faghri, M "Heat Trasnfer Memorial Award" Nov-2004
Papers from Meeting Proceedings	Charmchi, M.,Zhang, H. , Li, W. and Faghri, M "Solidification and Melting of Gallium in the Presence of Magnetic Field—Experimental Simulation of Low Gravity Environment" proceedings of 2004 ASME International Mechanical Engineering Congress &Exposition, Anaheim, CA. Nov-2004